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MILITARY SPECIFICATION

GYROSCOPE, DISPLACEMENT SBU-6/A, ROLL AND PITCH

1. SCOPE

1.1 This specification covers the requirements for one type of roll and pitch, displacement vertical gyroscope, designated SBU-6/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports, Preparation of
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-I-27680	Indicator, Attitude ARU-12/A, Remote
MIL-D-70327	Drawings, Engineering and Associated Lists

FSC 6615

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 MATERIALS

3.3.1 PROTECTIVE TREATMENT.- When materials are used in the construction of the gyroscope that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climate and environmental conditions shall be avoided.

3.3.2 METALS.- Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during normal service life. Hermetic sealing shall be considered adequate protection.

3.3.2.1 DISSIMILAR METALS.- Dissimilar metals are defined by MS33586. Unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided.

3.3.3 FUNGUS-PROOF MATERIALS.- Materials that are not nutrients for fungi shall be used to the greatest extent practicable. External materials that are nutrients for fungi shall be treated with a fungicidal agent, as approved by the procuring activity.

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3.3.4 CORROSIVE FUMES.- The materials, as installed in the gyroscope and under the service conditions specified in section 4, shall not liberate deleterious fumes.

3.4 DESIGN AND CONSTRUCTION.- The gyroscope shall be designed in accordance with figure 1 and as specified herein. It shall be built to withstand the temperature extremes, strains, jars, and vibration incident to shipping, storage, and installation. The gyroscope design shall include synchro outputs in roll and pitch.

3.4.1 The gyroscope shall be a conventional instrument having 2° of freedom with torque motors and synchros on each axis.

3.4.2 CONTROL.- Means shall be provided in the gyroscope for disconnecting roll or pitch erection, or both, through external switching devices.

3.4.3 MANUAL FAST ERECTION.- An external lead shall be provided for a manual push-button hold switch for application of fast erection to wash out small errors that may develop due to successive loops or other maneuvers.

3.5 PERFORMANCE.- The gyroscope shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Operating temperatures ranging from -54° to +71° C and storage temperatures ranging from -62° to +100° C

b. Pressures ranging from 30 inches Hg down to 0.82 inch Hg, approximately 80,000 feet altitude

c. Relative humidity up to 100 percent

d. Fungus growth as encountered in tropical climates

e. Rainfall as encountered in any locale

f. Exposure to salt sea atmosphere

g. Sand and dust particles as encountered in desert areas

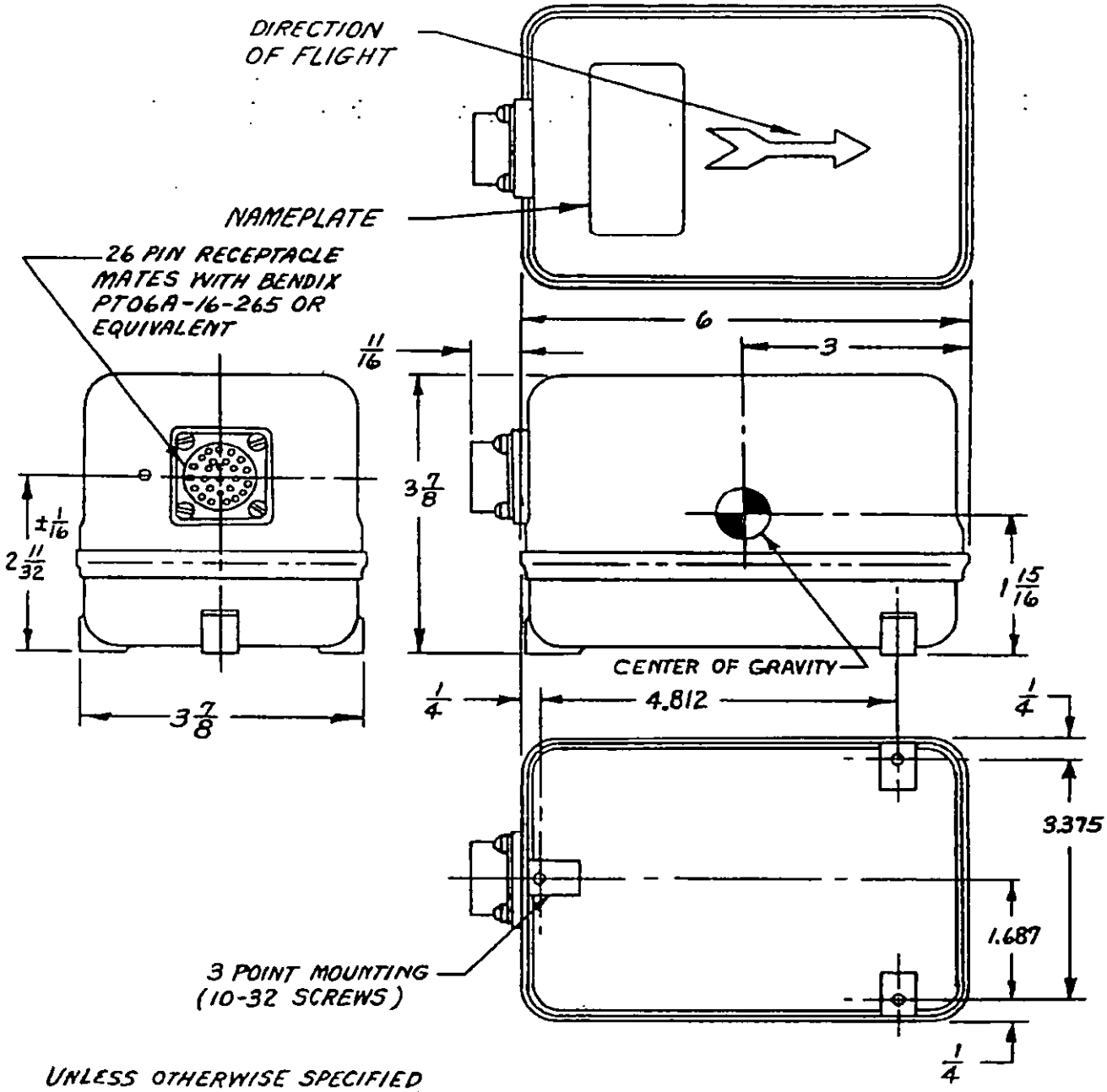
h. Vibration incident to service use

i. Shock forces as anticipated

j. A potential of 500V dc

k. Acceleration of 20g for 1 minute.

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UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS ±1/32
DECIMALS ±.005

FIGURE 1- VERTICAL GYRO

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3.5.1 RADIO INTERFERENCE.- The gyroscope shall meet the radio interference requirements of MIL-I-26600.

3.5.2 LIFE.- The gyroscope shall be capable of meeting the recuirements for 1,500 hours of operation during actual service use.

3.5.3 GIMBAL FREEDOM.- The gyroscope shall have unlimited freedom in roll and a minimum of ±82° of freedom in pitch. The pitch stops on the gyroscope or gimbal shall be so designed that no parts of the instrument will be damaged oy a loop at the rate of 90° per second, and less than ±3° error will result from loops of 10° ±1° per second. Errors induced by roll maneuvers at rates of 400° per second shall not exceed 1°.

3.5.4 MEAN-TIME-BETWEEN-FAILURES (MTBF).- The gyroscope shall have a MTBF of not less than 1,000 hours.

3.5.5 DRIFT.- Unless otherwise specified, the gyroscope shall be so designed that the maximum free drift, noted after compensation for the earth's rotation, will not be greater than 25° per hour when operated over the entire temperature range specified herein, with the gyro spin axis tilted 20° ±.50° or in the vertical position.

3.5.6 ERECTION.- Erection shall be accomplished by an electrolytic switch with a torque motor for each axis as shown on figure 2. When the vertical is displaced in one axis, it shall not deviate in the other axis by more than 1.0° while it is erecting. The erection system shall provide both initial and final erection and shall erect the gyroscope from any attitude to the vertical, as determined by the vertical sensing electrolytic switches.

3.5.6.1 INITIAL ERECTION.- The gyroscope shall have fast initial erection so that it assumes a position within 2° of vertical in pitch and bank within 68 seconds, or less, after power is applied. The rotor shall be at a standstill and the spin axis within the limits of ±85° bank and ±82° pitch from the vertical at the time the power is applied.

3.5.6.2 FINAL ERECTION.- The gyroscope shall align its spin axis with the direction of gravity within ±1/2° in 2 minutes, or less, after the initial erection period.

3.5.6.3 NORMAL ERECTION RATE.- The rate of erection in both the pitch and bank axes shall be between 1.5° and 3.0° per minute.

3.5.6.4 EXTERNAL TORQUING.- The gyroscope shall be so designed that it will precess off the vertical in both pitch and roll when certain pins on the electrical connector are externally connected. With pins K and H connected to pin J, the external torquing shall precess the top

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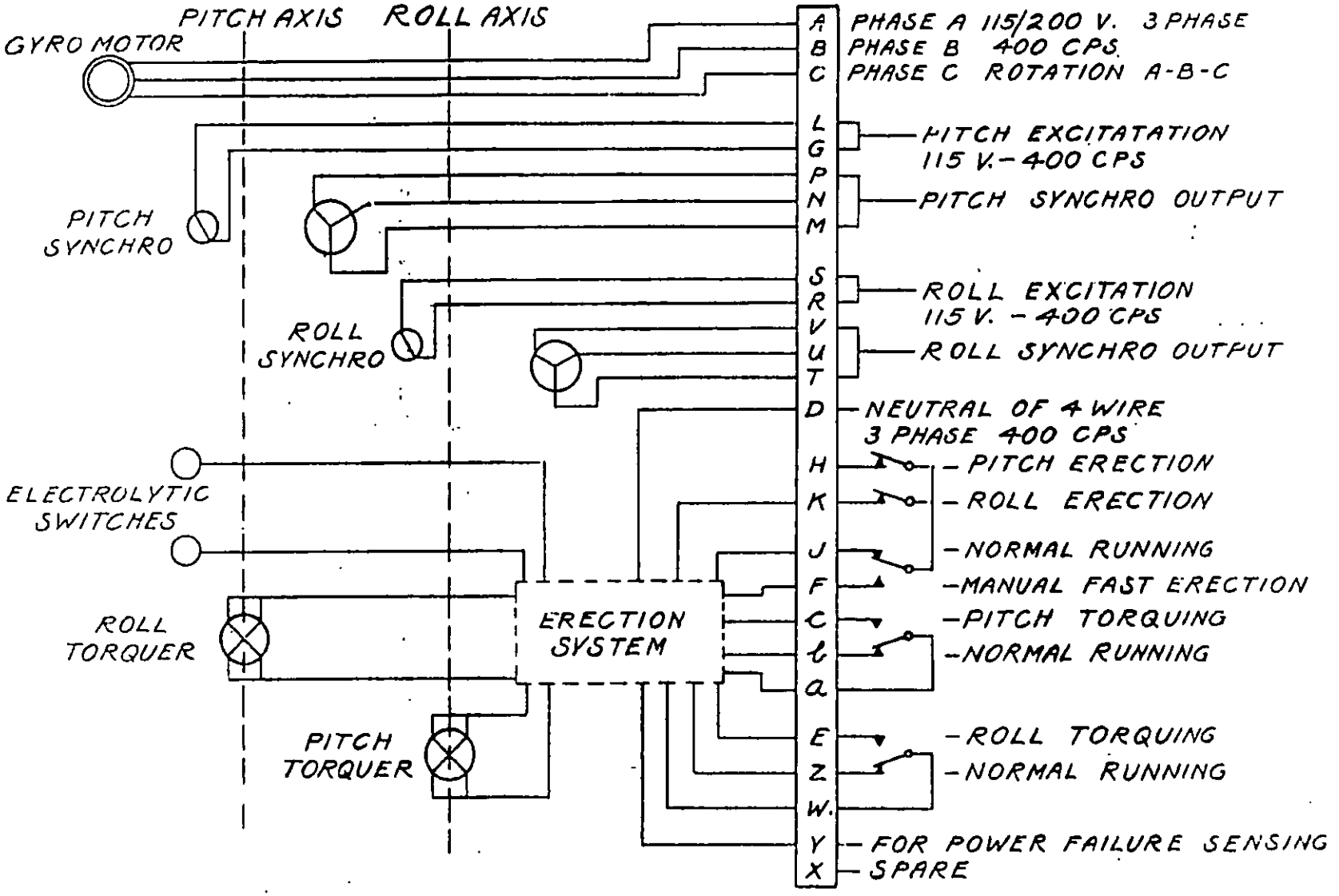


FIGURE 2 - VERTICAL GYRO SCHEMATIC

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of the gyro forward when pin a is disconnected from pin b and connected to pin c, and precess the top of the gyro to the right when pin W is disconnected from pin Z and connected to pin E (facing in the direction of the line-of-flight) at a rate of not less than 0.8° per minute. In addition, when pins K and H are disconnected from pin J and connected to pin F, the precession rate shall be not less than 15° per minute. Opening the circuit to pin H shall cause the gyro to cease to precess forward, and opening the circuit to pin H shall cause it to cease to precess to the right. The design shall be such that each axis may be precessed separately or both axes precessed simultaneously.

3.5.7 INVERTER SWITCHOVER DELAY.- Means shall be provided in the gyroscope to prevent system recycling through initial erection when power has been removed. After 5 minutes of operation at normal voltage, the gyroscope shall not recycle or develop errors for 30 seconds or less after power has been removed. Gyroscope gimbal freedom shall not be less than ±82° in pitch and ±60° in roll when power has been removed.

3.5.8 LEAK RATE.- The gyroscope shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to 80,000 feet altitude.

3.6 POWER FAILURE WARNING.- A 3-phase power warning circuit shall be incorporated in the gyroscope. This circuit shall operate a set of normally open contacts. Leads from this set of contacts shall be brought out of the case to pins Y and D as shown in table I. When proper power is applied to the gyroscope, the contact shall close the circuit between pins Y and D after the initial erection cycle is completed. If at any time any phase of the 3-phase power to the control is broken, the contacts shall open, breaking the circuit between pins Y and D.

3.7 PART NUMBERING OF INTERCHANGEABLE PARTS.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.8 PITCH AND ROLL OUTPUTS.- The electrical signals in pitch and roll shall be standard synchro type with a maximum nominal no-load of 11.8 $\pm 1/4$V line-to-line output and a leading phase shift of 4° ±1.5° through 360° roll and ±82° pitch maneuvers. Each synchro shall follow its respective gimbal movement at a 1 to 1 ratio.

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TABLE I

PIN CONNECTIONS

Pin	Function
A	115/200V, 400 cps
B	115/200V, 400 cps
C	115/200V, 400 cps
D	Neutral of 4-wire 3-phase Y, 400 cps
V, U, T	Roll synchro output
P, N, M	Pitch synchro output
Y	Power warning flag
H	Pitch erection cutout
K	Roll erection cutout
J	Normal erection running and starting (when connected to pins H and K)
F	Manual fast erection (when connected to pins H and K)
S, R	Roll synchro excitation (R is ground)
L, G	Pitch synchro excitation (G is ground)
a, b, c	Pitch external torquing (pin a connected to pin b for normal running and starting; pin a connected to pin c to precess top forward)
W, Z	Roll external torquing (pin W connected to pin Z its respective gimbal movement at a 1 to 1 ratio.

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3.9 SYNCHRO TRANSMITTER CHARACTERISTICS.- The transmitter shall be capable of operating up to five Bendix AY 500-5, or equal, control transformers. When loaded with five loads, the electrical characteristics of the transmitter shall not vary by more than the following:

a. The voltage drop from the nominal no-load line-to-line output shall not exceed 3.75V with the rotor leads of each control transformer open

b. The angle measured between the exciting voltage of each transmitter rotor and the voltage in any of the three stator leads of each of the five control transformers shall not exceed 3.5° increase in phase shift from the no-load value. This measurement shall be made with the rotor leads of the five control transformers open

c. Nulls measured up to five loads shall not exceed 0.03V. The conditions specified in a and b shall apply to the pitch and roll transmitters at room temperature.

3.10 SYNCHRO ZEROING AND SENSING.- The synchros shall be zeroed and their lead terminals, as defined with reference to the electrical receptacle pins shown in table I, shall be connected for the following procedures:

a. Connect terminals R and G to the ground side of a single-phase, 115V, 400-cps power source

b. Connect terminals S and L to the hot side of the power source

c. Connect terminals T and M to common point R and G. With the gyro gimbals at zero displacement about their axes, the voltage between V and U and between P and N shall each be at a null

d. With V and U strapped, the voltage between this point and S shall be less than 115V

e. With P and N strapped, the voltage between this point and L shall be less than 115V

f. With V and U unstrapped and the control case displaced in right bank about the roll axis with respect to the roll gimbal, the voltage between V and common point T and R shall increase before decreasing

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g. With P and N unstrapped and the control case displaced in climb about the pitch axis with respect to the pitch gimbal, the voltage between P and common point M and G shall increase before decreasing.

3.11 EXTERNAL CONNECTIONS.- When externally connected to the ARU-12/A indicator, conforming to MIL-I-27680, as shown on figure 3, the gyroscope shall operate as specified herein.

3.12 SYNCHRO EXCITATION.- The pitch and roll synchro rotor leads shall be isolated from the case or power leads and shall be made available at the connector. When the rotors are excited with phase A on pins S and L, and neutral on pins G and R, the outputs and sensing shall be in accordance with 3.8.

3.13 CASE.- The gyroscope case shall be hermetically sealed and of such strength that one atmosphere of differential pressure will not cause binding of either the pitch or roll gimbal.

3.13.1 FILLING MEDIUM.- The filling medium shall be 95 percent helium and 5 percent carbon dioxide. The gases used shall be at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65° C at filling pressure). The absolute pressure of the filling medium in the case shall be approximately 1 or 1/2 atmosphere, as required.

3.14 POWER.- The gyroscope shall operate from 200V, 3-phase, 400-cps Y power. The phase rotation shall be A, B, C from pins A to B to C. The gyroscope shall function satisfactorily with variations in voltage and frequency from 195 to 214V and 350 to 420 cps, respectively.

3.14.1 POWER CONSUMPTION.- The power consumption of the gyroscope shall not exceed 30 va after 5 minutes of operation and 55 va when fast erection is imposed at any time after 5 minutes of operation.

3.15 ELECTRONIC PARTS.- Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists or that do not conform to applicable Government specifications shall not be used unless approved by the procuring activity.

3.16 ELECTRICAL CONNECTOR.- A hermetically sealed connector shall be affixed to the gyroscope as shown on figure 1. The pin connections shall conform to table I.

3.17 DIMENSIONS.- The dimensions of the gyroscope shall be in accordance with figure 1.

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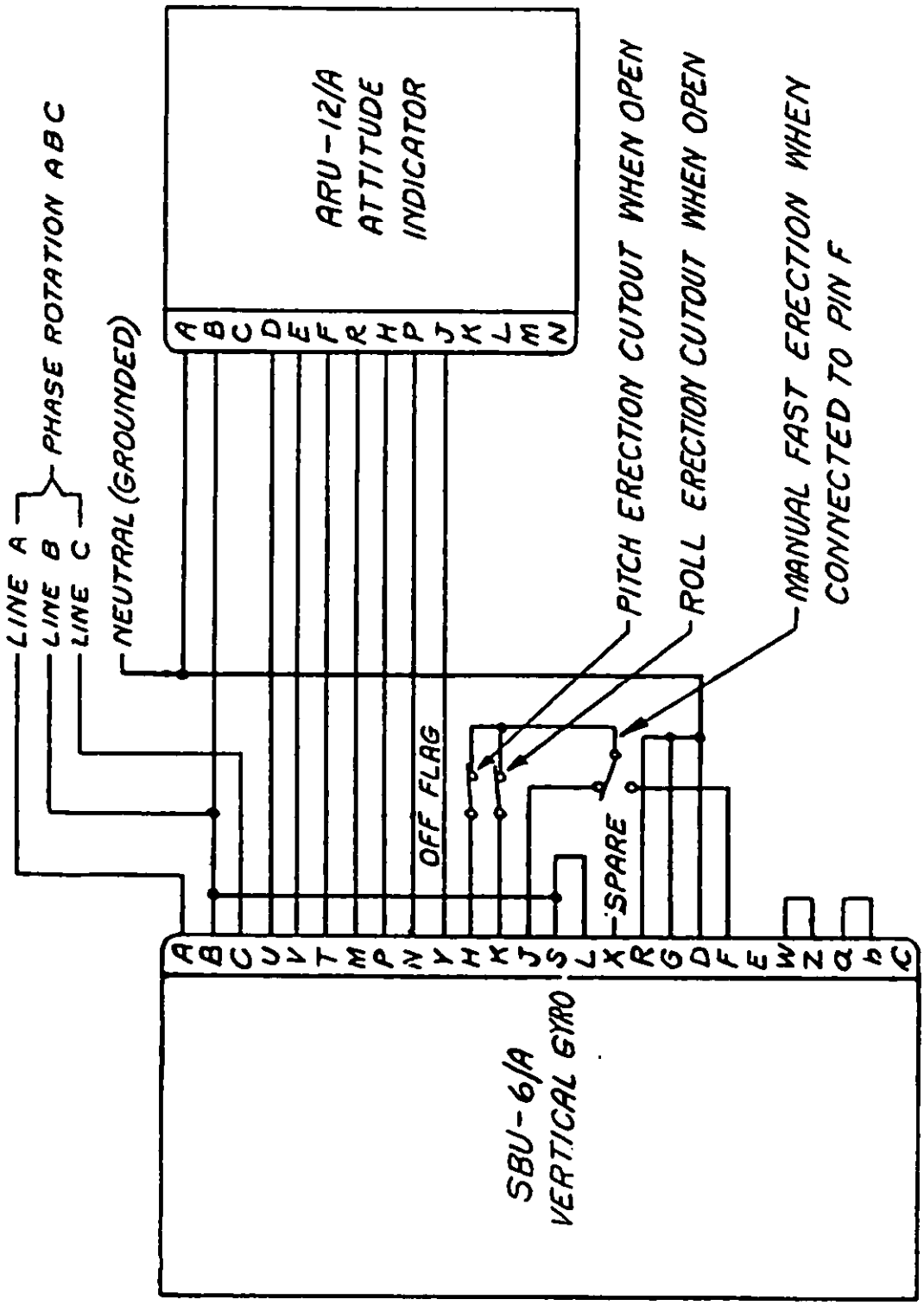


FIGURE 3- INTERCONNECT DIAGRAM

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3.18 WEIGHT.- The weight of the gyroscope shall not exceed 5.00 pounds.

3.19 CENTER OF GRAVITY.- The location of the center of gravity shall be in accordance with figure 1.

3.20 SOLDERING.- Soldering shall be accomplished in accordance with MIL-S-6872.

3.21 SCREW THREADS.- Unless otherwise specified, machine screw threads 0.060 inch diameter or larger shall conform to MIL-S-7742.

3.22 FINISHES AND PROTECTIVE COATINGS

3.22.1 ALUMINUM ALLOY PARTS.- Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Small holes and case inserts

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used

d. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541

e. Parts enclosed within a hermetically sealed container

f. Parts plated with tin over a copper flash

g. Where direct contact is necessary for electrical bonding.

3.22.2 STEEL PARTS.- Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.23 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.24 WORKMANSHIP.- The gyroscope, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 SUPPLIER'S RESPONSIBILITY.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of the gyroscope shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS.- Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches Hg, at a temperature of 25° ±5° C, and a relative humidity of 80 percent, or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.3.2 ATTITUDE.- Unless otherwise specified, the gyroscope shall be tested in its normal operating position.

4.3.3 VOLTAGE INPUT.- Unless otherwise specified, the gyroscope shall be tested with 200 ±2.5V, 3 phase, 400 ±5 cps. The phase rotation shall be as specified in 3.14.

4.3.4 OBSERVATION.- During testing, the position of the gimbals shall be determined from the outputs of the pitch and roll synchros. Suitable test equipment shall be used in all tests.

4.3.5 MASTER INDICATOR.- Unless otherwise specified, the gyroscope shall be connected to an indicator having an input impedance, as close as practicable, equivalent to the ARU-12/A attitude indicator conforming to MIL-I-27680.

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4.4 PREPRODUCTION TESTING

4.4.1 TEST SAMPLES.- Preproduction test samples shall consist of 3 gyroscopes representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The samples shall be prepared as follows:

4.4.1.1 DATA TO ACCOMPANY TEST SAMPLES.- The test samples shall be accompanied by the following data:

a. Brief operating data to enable test personnel to correctly operate the equipment

b. Assembly drawings (2 sets).

4.4.1.2 TEST REPORT, TEST SAMPLES, AND DATA.- When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Test report - Three copies of a test report in accordance with MIL-T-9107

b. Test samples - The samples which were tested plus one new untested sample

c. The data required under 4.4.1.1.

4.4.2 PREPRODUCTION TESTS.- The preproduction tests shall consist of all tests described under 4.6.

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests

4.5.1 INDIVIDUAL TESTS.- Each gyroscope shall be subjected to the following tests as described under 4.6 Test Methods:

a. Examination of product

b. Initial erection

c. Final erection

d. Power consumption

e. External torquing and erection rate

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f. Roll and pitch erection cutout

g. Manual fast erection

h. Drift.

i. Loop rate

j. Roll test

k. Inverter switchover delay

l. Gimbal freedom

m. Power warning circuit

n. Leak rate.

4.5.2 SAMPLING PLANS AND TESTS

4.5.2.1 SAMPLING PLAN A.- Two gyroscopes, selected at random from each 100 or less produced on the contract or order, shall be subjected to the following tests as described under 4.6 Test Methods:

a. Individual tests

b. Low voltage operation

c. Low temperature operation

d. High temperature operation

e. Dielectric strength

f. Vibration error.

4.5.2.2 SAMPLING PLAN B.- Unless otherwise specified, 3 gyroscopes shall be selected at random from the first 15 items on the contract or order and subjected to the following tests, in the sequence listed, as described under 4.6 Test Methods:

a. Sampling plan A tests

b. Radio noise interference

c. High temperature exposure

d. Low temperature-high altitude

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e. Humidity

f. Fungus

g. Salt spray

h. Sand and dust

i. Vibration failure

j. Shock

k. Acceleration

l. Thermal shock

m. Life

4.5.2.3 REJECTION AND RETEST.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, tests shall be repeated.

4.5.2.3.1 INDIVIDUAL TESTS MAY CONTINUE.- For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.2.3.2 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5.3 TEST GROUPING.- At the option of the procuring activity, the tests may be divided into 3 groups as follows, to be conducted on 3 gyroscopes, if a need exists to expedite tests. Each gyroscope shall meet the requirements of all the individual tests after completing the groups of special tests to which submitted:

Group I	Group II	Group III
Radio noise interference	Dielectric strength	Life
High temperature exposure	Humidity	Rain
Low temperature-high altitude	Fungus	Salt spray
Thermal shock		
Acceleration		
Vibration failure		

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4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT.- The gyroscope shall be inspected to determine compliance with the requirements specified herein with respect to materials, finish, marking, and workmanship.

4.6.2 INITIAL ERECTION.- A test table shall be levele within ±6 minutes. The gyroscope shall be placed on the test table with its spin axis at any random position within the limits of ±85° bank and ±82° pitch of its operating vertical position. Power shall then be applied. The gyro shall erect to within ±2° in pitch and bank by the end of the initial erection time delay period which shall not exceed 68 seconds. During this initial erection time delay period, there shall be no continuity between pins Y and D; however, at the end of the initial erection period, there shall be continuity between pins Y and D.

4.6.3 FINAL ERECTION.- A roll, pitch, and yaw mechanism and spin table shall be leveled within ±6 minutes. The gyroscope shall be allowed to erect for 2 minutes after the initial erection test. The gyroscope shall indicate level within ±1/2°. It shall then be rapidly rotated 180° in azimuth, and the pitch or bank axis shall not change by more than 1/2°. The low angle (not to exceed 2° total movement) roll, pitch, and yaw mechanism shall be operated at a speed of 5 to 7 oscillations per minute.

4.6.4 POWER CONSUMPTION.- After 5 minutes of operation, the motor power consumption shall not exceed 25 va.

4.6.5 EXTERNAL TORQUING AND ERECTION RATE

4.6.5.1 EXTERNAL TORQUING.- The gyroscope shall be mounted on a low angle (1° total movement) pitch and yaw mechanism. Power shall be applied and pin a connected to pin c and pin W to pin E. The top of the gyroscope shall precess forward and to the right (when facing the direction of the line-of-flight) at a rate of not less than 0.8° per minute when pins H and K are connected to pin J. Pins K and H shall be connected to pin F and the top of the gyroscope shall precess forward and to the right at a rate of not less than 15° per minute. The connection to pin H shall be removed and the top of the gyroscope shall cease to precess forward; the connection to pin K shall be removed and the top of the gyroscope shall cease to precess to the right. During this test, the synchro outputs from the gyroscope shall give a direct indication of gyroscope position at all times.

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4.6.5.2 ERECTION RATE.- The gyroscope shall be mounted on, and mechanically secured to, a leveled (within 1/2°) roll, pitch, and yaw mechanism oscillating simultaneously in three mutually perpendicular planes, with a total movement in each plane of 3° at a rate of 5 to 7 oscillations per minute. The power shall be applied to the gyroscope and time allowed for the gyroscope to reach operating speed. With the terminal end of the gyroscope oriented to either compass north or south, the gyroscope shall erect for 5° from an approximately 10° climb position at a rate of 1.5° to 2.5° per minute. During the erection tests, the undeflected axis (bank) shall not deviate from vertical more than 1.00°. The arithmetrical mean of dive and climb erection rates shall be 2.0 ±5° per minute. With the terminal end of the gyroscope oriented to either compass east or west, the gyro shall erect for 5° from an approximate simulated 10° right bank position at a rate of 1.5° to 2.5° per minute. The gyroscope shall also erect for 5° from an approximate simulated 10° left bank position at a rate within the above limits. During the erection tests, the undeflected axis (pitch) shall not deviate from vertical more than 1.00°. The arithmetical mean of the left and right bank erection rates shall be 2.0° ±.5 per minute.

4.6.6 ROLL AND PITCH ERECTION CUTOUT.- The gyroscope shall be deflected 10° in bank and 10° in pitch. Pins H and K shall be disconnected from pin J. There shall be no evidence of erection in either the roll or the pitch axis.

4.6.7 MANUAL FAST ERECTION.- With the gyroscope deflected as in 4.6.6, pins K and H shall be disconnected from pin J and connected to pin F. The gyroscope shall immediately start to erect at a minimum rate of 35° per minute.

4.6.8 DRIFT.- The gyroscope shall be mounted on a roll, pitch, and yaw mechanism with the connector end of the gyroscope pointed to the geographic west. The roll, pitch, and yaw mechanism shall be operated at a speed of 5 to 7 oscillations per minute and a total movement of 2.5°. With proper correction for the earth's rotation, the computed drift rate in either axis shall not exceed 25° per hour. The test shall be repeated with the gyroscope pointed to the south, east, and north, with the same tolerances applicable. The gyroscope shall be tested with the gyroscope spin axis tilted 20° east and with the undeflected axis level to within .5°; it may also be tested in its normal vertical position. If the drift in one heading exceeds the limit by not more than 1° in 5 minutes, the gyroscope may be retested in that heading and accepted if the first repeat test is within limits. Drift readings shall be recorded after a minimum period of 5 minutes at each heading.

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4.6.9 LOOP RATE.- The gyroscope shall be mounted on a fixture permitting rotation about a horizontal axis. The gyroscope shall be rotated one complete revolution (inside loop) at a rate of 10° ±1° per second, and returned to the normal position. The error in either pitch or bank shall not exceed 3°. The test mechanism shall be such that the axis not under test will be level within 0.50°.

4.6.10 ROLL TEST.- The gyroscope shall be mounted on a fixture rotating about a lateral axis. The gyroscope shall be rotated 5 complete revolutions (rolls) at a rate of 400° ±10° per second. The error induced in either pitch or bank shall not exceed 1°.

4.6.11 INVERTER SWITCHOVER DELAY.- After 5 minutes of operation, the power shall be removed and the gyroscope moved through angles of ±60° in pitch and ±55° in roll and returned to level. Power shall be applied 30 seconds after removal, and the pitch and bank axes shall indicate level within ±1°.

4.6.12 GIMBAL FREEDOM.- With power applied to the gyroscope, it shall be moved 360° in bank and ±82° in pitch. There shall be no indication of gimbal sticking or binding in either axis.

4.6.13 POWER WARNING CIRCUIT.- The gyroscope shall be properly connected and power applied. The external connection to pin A shall be removed and the circuit between pins Y and D shall open immediately after the gyroscope has reached full speed. Pin A shall be connected and the circuit between pins X and D shall close. This procedure shall be repeated with pins B and C. The same results shall occur as with pin A.

4.6.14 LEAK RATE.- The sealed case shall be tested for leaks using a helium leak detector or other means approved by the procuring activity. The leak rate so determined shall not be sufficient to cause more than 10 percent loss or contamination of the filling medium after 1,000 hours at approximately 1 atmosphere differential pressure.

4.6.15 LOW VOLTAGE OPERATION.- Power shall be applied to the gyroscope and tests run at the following combinations of voltage and frequency. The gyroscope shall then meet the tests specified in 4.6.8, 4.6.12, and 4.6.13:

a. 195V 350 cps

b. 195V 420 cps

c. 214V 350 cps

d. 214V 420 cps

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4.6.16 LOW TEMPERATURE OPERATION.- The gyroscope shall be placed in a chamber at a temperature of -54° ±2° C for a period of 4 hours. At the end of this period and while at the reduced temperature, the gyroscope shall be subjected to and shall meet the tests specified in 4.6.2, 4.6.3, 4.6.8, and 4.6.12 with the following exceptions:

a. Before conducting the initial erection test, the spin axis shall be displaced at any random pitch angle within ±82° and any random roll angle within ±60° from the vertical at the time power is applied. The initial erection time shall not exceed 1-1/2 minutes after power is applied.

b. Power shall be applied for 10 minutes before conducting the final erection test.

c. The erection rate limits shall be 1.5° to 3.0° per minute with a 3° maximum deviation on the undeflected axis.

d. The no-load output shall be 12.2 ±0.3V and the no-load phase shall be 3.5° ±1.5°. The voltage drop from the nominal no-load output shall not exceed 1.2V for the five loads (rotors open). The phase shift with five loads (rotors open) shall not exceed a 2.5° increase in phase shift from the no-load value.

4.6.17 GIMBAL CLEARANCE.- Three gyroscopes that have passed the test specified in 4.6.12 shall be subjected to a differential pressure equivalent to that between sea level and 80,000 feet. The outside pressure on the case shall be 29.92 inches Hg and the inside pressure shall be equivalent to 80,000 feet. At the conclusion of this test, the gyroscope shall be subjected to the test specified in 4.6.12.

4.6.18 HIGH TEMPERATURE OPERATION.- The gyroscope shall be properly connected and placed in a chamber at a temperature of +71° ±2° C for a period of 4 hours. The gyroscope shall be operated during the entire test. Immediately after the 4-hour period and while at the high temperature, the gyroscope shall be subjected to the tests specified in 4.6.2, 4.6.3, and 4.6.8, except for the following synchro transmitter condition:

a. The no-load output shall be 11.5 ±0.3V and the no-load phase shall be 6.6° ±2.5°. The voltage drop from the nominal no-load output shall not exceed 1.2V for the five loads (rotor open). The phase shift with the five loads (rotors open) shall not exceed a 4.5° increase in phase shift from the no-load value.

4.6.18.1 Using suitable test equipment, the synchro transmitter electrical characteristics specified in 3.9 shall be measured and shall be within the tolerances specified therein.

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4.6.19 DIELECTRIC STRENGTH.- A potential of 500V dc shall be applied between isolated pins and between pins and the case. The measured insulation resistance shall be greater than 50 megohms, and there shall be no insulation breakdown. This test shall not be made in circuits connected with the gyroscope erection switch.

4.6.20 VIBRATION ERROR.- With the gyroscope operating, it shall be vibrated along the 3 perpendicular axes with a complex wave vibration. The input power spectral density shall be $0.002g^2/cps$ for each frequency from 20 to 2,000 cps. The signal source shall be either a prepared magnetic tape or a noise generator. The duration of the test shall be 15 minutes. After vibration, the unit shall be checked as stated in 4.6.2, 4.6.3, 4.6.5.1, 4.6.8, and 4.6.12. (Note: Proper compensation for the leading effect on the test equipment shall be made to maintain the power spectrum density.)

4.6.20.1 SIGNAL CONTINUITY.- There shall be no transients or discontinuity in the pitch and roll synchro outputs during vibration.

4.6.20.2 VIBRATION.- After 1 minute of vibration, the pitch and roll axes shall not vary by more than 1° from their original position before vibration.

4.6.21 RADIO NOISE INTERFERENCE.- The gyroscope shall be subjected to radio noise interference tests in accordance with MIL-I-26600.

4.6.22 HIGH TEMPERATURE EXPOSURE.- With the gyroscope properly connected, it shall be subjected to a high temperature exposure test in accordance with MIL-E-5272, procedure II. At the end of the 48-hour period and while still at the high temperature, power shall be applied and the gyroscope subjected to the tests specified in 4.6.2, 4.6.3, 4.6.8, and 4.6.12. The temperature shall be raised to 100° ±2° C within a 15-minute period and maintained for 30 minutes. The temperature shall be reduced to 71° ±2° C within a 15-minute period and maintained for 2 hours and 30 minutes. At the end of this period, the gyroscope shall be subjected to the test specified in 4.6.12.

4.6.23 LOW TEMPERATURE-HIGH ALTITUDE.- The gyroscope shall be subjected to a low temperature-high altitude test in accordance with MIL-E-5272, procedure II. The chamber shall be reduced to -62° C and the gyroscope properly connected but no power applied. After a period of 72 hours, the pressure in the chamber shall be reduced to the equivalent of 80,000 ±500 feet and power applied. The time required for initial erection shall not exceed 5 minutes, and there shall be no indication of gimbal sticking or binding in either axis. After the gyroscope has returned to room temperature, it shall be subjected to the individual tests. In the initial erection test, the spin axis shall be displaced at any random pitch angle within ±82° and at any random roll angle within ±60° from vertical at the time power is applied.

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4.6.24 HUMIDITY.- The gyroscope shall be subjected to a humidity test in accordance with MIL-E-5272, procedure I, after which it shall meet the test specified in 4.6.12. There shall be no evidence of corrosion or rusting of external parts that would adversely affect subsequent operation of the gyroscope.

4.6.25 FUNGUS.- The gyroscope shall be subjected to a fungus test in accordance with MIL-E-5272, procedure I, after which it shall meet the test specified in 4.6.12. There shall be no deterioration that would adversely affect subsequent operation nor shall any external part support fungus growth.

4.6.26 RAIN.- The gyroscope, mounted in its normal operating position, shall be subjected to a rain test in accordance with MIL-E-5272, procedure II, after which it shall meet the tests specified in 4.6.3, 4.6.4, and 4.6.12.

4.6.27 SALT SPRAY.- The gyroscope shall be subjected to the salt spray test of MIL-E-5272, procedure I. At the end of the 50-hour period, the gyroscope shall be subjected to the individual tests.

4.6.28 SAND AND DUST.- The gyroscope shall be subjected to a sand and dust test in accordance with procedure I of MIL-E-5272.

4.6.29 VIBRATION FAILURE.- The gyroscope, properly connected and with power applied, shall be subjected to the vibration test of MIL-E-5272, procedure V, except that the circular motion shall be 0.028 to 0.030 inch diameter. At the end of this test, the gyroscope shall be subjected to the individual tests.

4.6.30 SHOCK.- The gyroscope shall be subjected to a shock test in accordance with procedure V of MIL-E-5272.

4.6.31 ACCELERATION.- The gyroscope, not operating, shall be mounted on a centrifuge in its normal manner and subjected to an acceleration of 20g along its vertical axis and then along each of the two axes that are perpendicular to the vertical axis and to each other for a period of 1 minute. At the end of this test, the gyroscope shall be subjected to the individual tests. No damage to the gyroscope shall result.

4.6.32 THERMAL SHOCK.- The gyroscope shall be subjected to a temperature shock test in accordance with MIL-E-5272, procedure I. After the gyroscope has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or seal leakage. The gyroscope shall then be subjected to the test specified in 4.6.8.

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4.6.33 LIFE.- The gyroscope shall be properly connected and operated for 1,500 hours on a 15° total movement roll, pitch, and yaw mechanism having 5 to 7 oscillations per minute. There shall be no failure during this test. This test may be performed as a 1,500-hour continuous life test without interruption, or a 1,500-hour test of cycling with the gyroscope operating continuously for a 22-hour period, then with the gyroscope off for a 2-hour period. The test time shall then be computed from the summation of the 22-hour operating periods. Upon completion of the test, the gyroscope shall meet the individual tests.

4.6.34 MEAN-TIME-BETWEEN-FAILURES.- Upon completion of all tests specified under 4.6, a MTBF shall be calculated, exclusive of group II tests. The MTBF shall be not less than 1,000 hours. In the event an MTBF of 1,000 hours cannot be verified, three production units shall be selected at random and subjected to the test specified in 4.6.33. The calculated MTBF for the retest samples shall be not less than 1,500 hours. In the event 1,500 hours MTBF cannot be established, a design evaluation shall be made and necessary action taken. MTBF shall be considered as the total operating time accumulated on all test units divided by the number of failures. All out-of-tolerance conditions shall be cause for rejection. For the purpose of this test, out-of-tolerance conditions are defined as follows:

a. In those paragraphs which specify a tolerance-type requirement, a failure shall be considered as an electrical or mechanical condition which results in an increase of 100 percent or more of the specified tolerance.

b. In those paragraphs which specify a maximum- or minimum-type requirement, a failure shall be considered as any electrical or mechanical condition which results in a 10 percent deviation from the specified maximum or minimum.

c. In those paragraphs where a condition is specified, a failure shall be considered as any electrical or mechanical malfunction that results in any condition other than that specified.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION, PACKAGING, AND PACKING

5.1.1 The gyroscope shall be preserved in accordance with MIL-P-116, method II, and packaged and packed in accordance with MIL-P-7936.

5.2 MARKING OF SHIPMENTS.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 INTENDED USE.- The SBU-6/A gyroscope covered by this specification is intended for use as a vertical reference in missile and drone applications, and in manned aircraft where maximum performance under severe environmental conditions is required.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. If sampling plan B tests are not to be conducted.

c. Selection of applicable levels of packaging and packing.

6.3 DEFINITION

6.3.1 HERMETIC SEAL.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or between vitric and metallic materials. A hermetic seal is not intended to include seals accomplished by gaskets.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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